

Gravitation, solved

Step-by-step solutions to every practice problem in Chapter 8. Try each problem on your own before you read its solution.

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P8.1

The Moon is falling towards the Earth, continuously. Its acceleration towards the Earth is the centripetal acceleration of its orbit. It also has a sideways velocity of about 1.02 km/s. In each second it falls a little way towards the Earth from the straight line it would otherwise follow, and that fall exactly matches the curvature of its circular path. So it keeps falling *around* the Earth without getting any closer.

Newton’s “Moon test”. The Moon’s centripetal acceleration is

$$a = \frac{v^2}{r} = \frac{(1022)^2}{3.84 \times 10^8} = 2.72 \times 10^{-3} \text{ m/s}^2.$$

If gravity falls off as $1/r^2$, the Moon (about 60.3 Earth radii away) should accelerate at

$$g \left(\frac{1}{60.3} \right)^2 = 9.8 \times 2.75 \times 10^{-4} = 2.70 \times 10^{-3} \text{ m/s}^2.$$

The two agree. This was Newton’s evidence that the same gravity governs apples and moons. In each second the Moon “falls” $\frac{1}{2}at^2 = 1.4 \text{ mm}$.

P8.2

At the ISS’s altitude, gravity is about 89% of its surface value, so it is far from zero. If gravity really switched off, the station would fly off in a straight line.

A better description: the station and the astronauts are in **free fall**. They all accelerate towards the Earth at the same rate. Nothing supports the astronauts relative to the station, so their *apparent weight* is zero. Terms such as “microgravity” or “free fall” are more accurate than “zero gravity”.

P8.3

1. **Rotation.** At the equator, part of the gravitational force is needed to provide the centripetal acceleration $\omega^2 R = (7.29 \times 10^{-5})^2 (6.38 \times 10^6) = 0.034 \text{ m/s}^2$ for circling the Earth's axis once a day. The *effective* g measured by a scale is reduced by this amount. At the poles, the centripetal acceleration is zero.
2. **Shape.** The Earth bulges at the equator, partly because of its rotation, with an equatorial radius about 21 km larger than the polar radius. A point on the equator is farther from the centre, so GM/R^2 is slightly smaller there.

P8.4

It speeds up. For a circular orbit, $K = -E$ and $U = 2E$. Drag removes energy, so E becomes more negative. Then $K = -E$ increases, while $U = 2E$ decreases twice as fast.

How it works. Drag acts against the motion and lowers the orbit slightly. The satellite drifts inward along a gentle spiral. On that spiral, its velocity has a small component *towards* the Earth, so gravity does positive work on it. That work is twice as large as the negative work done by drag. The net effect is that the satellite moves lower and faster. Low satellites in orbits that decay this way speed up as they fall, until the drag becomes so large that they re-enter the atmosphere.

P8.5

$$g_{\text{Mars}} = \frac{GM}{R^2} = \frac{(6.674 \times 10^{-11})(6.42 \times 10^{23})}{(3.39 \times 10^6)^2} = \frac{4.28 \times 10^{13}}{1.149 \times 10^{13}} = 3.73 \text{ m/s}^2.$$

That is about 38% of Earth's.

Weight of the astronaut plus suit: $(80 + 40)(3.73) = 448 \text{ N}$. On Earth the same 120 kg would weigh 1180 N, so even with a heavy suit the astronaut feels lighter than on Earth without one.

P8.6

We need $\left(\frac{R}{R+h}\right)^2 = \frac{1}{2}$, so

$$R+h = \sqrt{2}R \Rightarrow h = (\sqrt{2}-1)R = 0.414R = 0.414(6371 \text{ km}) = 2640 \text{ km}.$$

P8.7

Each pair. Every pair of spheres attracts with the same force:

$$F = \frac{Gm^2}{d^2} = \frac{(6.674 \times 10^{-11})(2.0)^2}{1.0^2} = 2.67 \times 10^{-10} \text{ N.}$$

Net force on one sphere. The two forces on it are 60° apart. Each makes 30° with the bisector of the angle at that corner. Their components perpendicular to the bisector cancel, and their components along it add:

$$F_{\text{net}} = 2F \cos 30^\circ = 2(2.67 \times 10^{-10})(0.866) = 4.6 \times 10^{-10} \text{ N,}$$

directed towards the centre of the triangle, by symmetry.

P8.8

At the balance point, a distance x from the Earth's centre:

$$\frac{GM_E m}{x^2} = \frac{GM_M m}{(d-x)^2} \Rightarrow \frac{x}{d-x} = \sqrt{\frac{M_E}{M_M}} = \sqrt{81.3} = 9.02.$$

$$x = \frac{9.02}{10.02}d = 0.900d = 0.900(3.84 \times 10^8) = 3.46 \times 10^8 \text{ m.}$$

The point is 90% of the way to the Moon, about 38 400 km from the Moon's centre.

(Strictly, in the rotating Earth–Moon frame the true equilibrium point, the Lagrange point L1, also includes the centrifugal force and lies at about 326 000 km from the Earth. The simple calculation asked for here ignores the rotation.)

P8.9

$$r = 6.371 \times 10^6 + 2.0 \times 10^6 = 8.371 \times 10^6 \text{ m.}$$

$$v = \sqrt{\frac{GM}{r}} = \sqrt{\frac{3.986 \times 10^{14}}{8.371 \times 10^6}} = \sqrt{4.76 \times 10^7} = 6.90 \text{ km/s.}$$

$$T = \frac{2\pi r}{v} = \frac{2\pi(8.371 \times 10^6)}{6900} = 7.62 \times 10^3 \text{ s} = 127 \text{ min.}$$

P8.10(a) **Surface gravity:**

$$g = \frac{GM}{R^2} = \frac{(6.674 \times 10^{-11})(2.8 \times 10^{30})}{(1.0 \times 10^4)^2} = 1.9 \times 10^{12} \text{ m/s}^2.$$

That is about 2×10^{11} times the Earth's.

(b) **Escape speed:**

$$v_{\text{esc}} = \sqrt{\frac{2GM}{R}} = \sqrt{\frac{2(6.674 \times 10^{-11})(2.8 \times 10^{30})}{1.0 \times 10^4}} = \sqrt{3.74 \times 10^{16}} = 1.93 \times 10^8 \text{ m/s} = 0.64 c.$$

Evaluate. At such speeds Newtonian gravity is only approximate, and general relativity is needed. Compressing the star by another factor of about 3 would push the escape speed past c and make it a black hole (Problem P8.18).

P8.11**Radius,** from Kepler's third law:

$$r^3 = \frac{GMT^2}{4\pi^2} = \frac{(3.986 \times 10^{14})(43\,082)^2}{39.48} = \frac{7.398 \times 10^{23}}{39.48} = 1.874 \times 10^{22} \text{ m}^3.$$

$$r = 2.66 \times 10^7 \text{ m}.$$

Altitude: $2.656 \times 10^7 - 6.371 \times 10^6 = 2.02 \times 10^7 \text{ m}$, about 20 200 km.

Speed:

$$v = \sqrt{\frac{GM}{r}} = \sqrt{\frac{3.986 \times 10^{14}}{2.656 \times 10^7}} = 3.87 \text{ km/s}.$$

Check with Kepler's third law in ratio form. The GPS period is half the geostationary period, so its radius should be $(1/2)^{2/3} = 0.63$ times the geostationary radius: $0.63 \times 4.22 \times 10^7 = 2.66 \times 10^7 \text{ m}$. ✓

P8.12

(a) **Rotation rate.** At the rim we need $\omega^2 r = g$:

$$\omega = \sqrt{\frac{g}{r}} = \sqrt{\frac{9.8}{100}} = 0.313 \text{ rad/s} = 0.313 \times \frac{60}{2\pi} = 3.0 \text{ rpm.}$$

(b) **Head-to-foot gradient.** The apparent gravity is proportional to r . The feet are at 100 m and the head at 98 m, so the head feels

$$\frac{98}{100} = 0.98$$

of the gravity at the feet: 2% less. That is too small to notice, unlike the gradient in a small, fast-spinning centrifuge.

(c) **Why the rotation rate matters.** In a rotating frame, anything that *moves*, including the fluid in the semicircular canals of the inner ear, feels Coriolis forces. Turning or nodding your head while the frame rotates excites the canals in an unnatural “cross-coupled” way. This causes disorientation and nausea, and the effects become troublesome above about 2–4 rpm. Since $\omega = \sqrt{g/r}$, a larger radius gives the same artificial gravity at a lower spin rate. For 1 g at 2 rpm (0.209 rad/s), the radius must be $r = g/\omega^2 = 9.8/0.0438 = 224$ m. That is a very large structure, which is one reason no such station has yet been built.

P8.13

(a) **Force inside.** At distance r from the centre, only the mass inside radius r attracts (shell theorem). For a uniform Earth that mass is $M(r/R)^3$:

$$F = -\frac{G M(r/R)^3 m}{r^2} = -\frac{GMm}{R^3} r = -\frac{mg}{R} r,$$

using $g = GM/R^2$.

(b) **Simple harmonic motion.** The force is a linear restoring force, $F = -kr$ with $k = mg/R$, which is the condition for SHM (Chapter 9). The angular frequency and period are

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{g}{R}}, \quad T = 2\pi\sqrt{\frac{R}{g}} = 2\pi\sqrt{\frac{6.371 \times 10^6}{9.8}} = 2\pi(806) = 5.07 \times 10^3 \text{ s} = 84 \text{ min.}$$

A one-way trip to the other side of the Earth takes 42 min.

(c) **Maximum speed**, at the centre:

$$v_{\max} = \omega R = \sqrt{gR} = \sqrt{9.8(6.371 \times 10^6)} = 7.9 \text{ km/s.}$$

(d) **The coincidence.** A satellite orbiting just above the surface has $T = 2\pi\sqrt{R^3/GM} = 2\pi\sqrt{R/g}$, which is **the same** 84 min, and its speed is also $\sqrt{gR}$. This is no accident. The projection of uniform circular motion onto a diameter is SHM. The capsule moves exactly like the “shadow” of the satellite on the tunnel. In fact, *any* straight frictionless tunnel through a uniform Earth, not only one through the centre, gives the same 42 min one-way trip.

P8.14

Use $GM = 3.986 \times 10^{14} \text{ m}^3/\text{s}^2$. The transfer ellipse has semi-major axis

$$a = \frac{r_1 + r_2}{2} = \frac{6.671 \times 10^6 + 4.22 \times 10^7}{2} = 2.444 \times 10^7 \text{ m.}$$

Circular speeds:

$$v_1 = \sqrt{\frac{GM}{r_1}} = 7.73 \text{ km/s}, \quad v_2 = \sqrt{\frac{GM}{r_2}} = 3.07 \text{ km/s.}$$

(a) **At perigee of the transfer orbit**, from vis-viva:

$$v_p = \sqrt{GM \left(\frac{2}{r_1} - \frac{1}{a} \right)} = \sqrt{3.986 \times 10^{14} (2.998 \times 10^{-7} - 4.09 \times 10^{-8})} = 10.16 \text{ km/s.}$$

$$\Delta v_1 = 10.16 - 7.73 = 2.43 \text{ km/s.}$$

(b) **At apogee**, by conservation of angular momentum:

$$v_a = v_p \frac{r_1}{r_2} = 10.16 \times \frac{6.671}{42.2} = 1.61 \text{ km/s.}$$

To circularise at r_2 :

$$\Delta v_2 = 3.07 - 1.61 = 1.47 \text{ km/s.}$$

(c) **Total:** $\Delta v = 2.43 + 1.47 = 3.90 \text{ km/s.}$

(d) **Time on the transfer orbit** = half the period of the ellipse:

$$t = \pi \sqrt{\frac{a^3}{GM}} = \pi \sqrt{\frac{(2.444 \times 10^7)^3}{3.986 \times 10^{14}}} = \pi(6050) = 1.90 \times 10^4 \text{ s} = 5.3 \text{ h.}$$

Evaluate. Both burns are made in the direction of motion, yet the satellite ends up moving *slower* (3.07 against 7.73 km/s). Between the burns, gravity does a large amount of negative work as the satellite climbs from perigee to apogee.

P8.15

(a) **Energy conservation**, from rest at $2R$ to the surface at R :

$$0 - \frac{GMm}{2R} = \frac{1}{2}mv^2 - \frac{GMm}{R} \Rightarrow v^2 = \frac{GM}{R} = gR.$$

$$v = \sqrt{gR} = \sqrt{9.8(6.371 \times 10^6)} = 7.9 \text{ km/s.}$$

(b) **The naive formula.** $\sqrt{2gh}$ with $h = R$ gives $\sqrt{2gR} = 11.2 \text{ km/s}$, which is 41% too high. It assumes that gravity is g all the way up, but at $2R$ gravity is only $g/4$. The real average pull over the fall is weaker, so less work is done.

(c) **From infinity:**

$$\frac{1}{2}mv^2 = \frac{GMm}{R} \Rightarrow v = \sqrt{2gR} = 11.2 \text{ km/s,}$$

the escape speed. By energy conservation, anything falling from far away arrives at least this fast. That is why meteoroids hit the atmosphere at 11–72 km/s.

P8.16

(a) **Set up.** Each star moves on a circle of radius $d/2$ about the centre of mass. The gravitational force between them provides the centripetal force:

$$\frac{GM^2}{d^2} = \frac{Mv^2}{d/2} \Rightarrow v^2 = \frac{GM}{2d}.$$

$$T = \frac{2\pi(d/2)}{v} = \pi d \sqrt{\frac{2d}{GM}} = 2\pi \sqrt{\frac{d^3}{2GM}}. \checkmark$$

This is Kepler's third law with the *total* mass $2M$ in place of M .

(b) **Numbers:**

$$T = 2\pi \sqrt{\frac{(1.496 \times 10^{11})^3}{2(6.674 \times 10^{-11})(1.99 \times 10^{30})}} = 2\pi \sqrt{1.26 \times 10^{13}} = 2.23 \times 10^7 \text{ s} = 0.71 \text{ yr.}$$

This is exactly $1/\sqrt{2}$ of a year: doubling the total mass at the same separation shortens the period by a factor of $\sqrt{2}$.

Evaluate. Measuring the periods and separations of binary stars is the main way astronomers measure stellar masses.

P8.17

(a) **Derivation.** The acceleration is $a(d) = GM/d^2$. For a small change Δr in distance:

$$\Delta a \approx \left| \frac{da}{dd} \right| \Delta r = \frac{2GM}{d^3} \Delta r.$$

(b) **Moon:**

$$\Delta a_M = \frac{2(6.674 \times 10^{-11})(7.35 \times 10^{22})(6.371 \times 10^6)}{(3.84 \times 10^8)^3} = \frac{6.25 \times 10^{19}}{5.66 \times 10^{25}} = 1.10 \times 10^{-6} \text{ m/s}^2.$$

Sun:

$$\Delta a_S = \frac{2(6.674 \times 10^{-11})(1.99 \times 10^{30})(6.371 \times 10^6)}{(1.496 \times 10^{11})^3} = \frac{1.69 \times 10^{27}}{3.35 \times 10^{33}} = 5.1 \times 10^{-7} \text{ m/s}^2.$$

The Moon's tidal effect is $1.10/0.51 = 2.2$ times the Sun's.

(c) **Spring and neap tides.** At new and full moon, the Sun, Earth and Moon lie in a line, and the solar and lunar bulges add. That gives the largest tidal range, *spring tides* (nothing to do with the season). At first and last quarter, the Sun and Moon pull at right angles. The solar bulge then partly fills in the lunar low tides, giving the smallest range, *neap tides*. Their ratio is roughly $(1.10 + 0.51)/(1.10 - 0.51) \approx 2.7$.

P8.18

(a) **Schwarzschild radius.** Set $v_{\text{esc}} = \sqrt{2GM/R} = c$:

$$R_s = \frac{2GM}{c^2}.$$

- Sun: $R_s = \frac{2(6.674 \times 10^{-11})(1.99 \times 10^{30})}{(3.0 \times 10^8)^2} = 2.95 \times 10^3$ m, about 3 km. Its actual radius is 696 000 km. - Earth: $R_s = \frac{2(3.986 \times 10^{14})}{9.0 \times 10^{16}} = 8.9 \times 10^{-3}$ m, about 9 mm, the size of a marble.

(b) Tidal stretching at R_s . Using $2GM = c^2 R_s$:

$$\Delta a = \frac{2GM}{R_s^3} h = \frac{c^2 h}{R_s^2}.$$

(i) A 10-solar-mass black hole has $R_s = 29.5$ km:

$$\Delta a = \frac{(9.0 \times 10^{16})(2.0)}{(2.95 \times 10^4)^2} = 2.1 \times 10^8 \text{ m/s}^2 \approx 2 \times 10^7 g.$$

The astronaut would be torn apart long before reaching R_s (“spaghettification”).

(ii) A 4×10^6 -solar-mass black hole has $R_s = 4 \times 10^6 \times 2.95 \text{ km} = 1.18 \times 10^{10}$ m:

$$\Delta a = \frac{1.8 \times 10^{17}}{(1.18 \times 10^{10})^2} = 1.3 \times 10^{-3} \text{ m/s}^2.$$

This would be imperceptible.

Conclusion. Paradoxically, an astronaut could cross the event horizon of a *supermassive* black hole without feeling anything unusual, because the tidal acceleration scales as $1/M^2$ at the horizon. Bigger black holes are gentler at their edges.

All worked examples and practice problems in these notes are original.

Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

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